

AHMAD DZAKI

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PROFESSIONAL SUMMARY

Fresh graduate (D4) in Electronics Engineering (Majoring Telecommunications) from Politeknik Caltex Riau (GPA 3.66/4.00) with hands-on experience in embedded systems, IoT, RF/antenna design (CST Studio Suite), and computer networking, gained through two engineering internships and multiple independent projects. Skilled in hardware-software integration, instrumentation, troubleshooting, and root-cause analysis, with a proven ability to coordinate technical work across cross-functional teams. Actively seeking opportunities in electronics, embedded systems/IoT, RF/EMC, or technical field support roles.

EDUCATION

Applied Bachelor's Degree (D4), Electronics Engineering (Majoring Telecommunications)

Politeknik Caltex Riau — Pekanbaru, Riau, Indonesia | Aug 2022 – Aug 2026

- GPA: 3.66 / 4.00.
- Relevant coursework: electronic circuit fundamentals, digital & analog systems, embedded systems, Internet of Things (IoT), industrial automation fundamentals, instrumentation, Electromagnetic Compatibility (EMC), wireless and fiber optic communication, computer networking.
- Recipient of the Riau Provincial Government Merit Scholarship, 2023–2026.

CORE COMPETENCIES

Electronics & EMC/RF: Electronic circuit fundamentals (analog & digital), Electromagnetic Compatibility (EMC) fundamentals, RF & antenna design/simulation (CST Studio Suite), instrumentation & measurement.

Embedded Systems & IoT: Microcontrollers (ESP32, Arduino), sensor integration, RFID systems, GPS tracking, hardware-software integration, data acquisition.

Networking & IT: LAN troubleshooting & maintenance, network infrastructure monitoring, Cisco Packet Tracer, CCNA fundamentals.

Software Development: Web application development (front-end & back-end) using PHP, MySQL database, Google App Script.

Programming: C/C++ (embedded systems), basic Python.

Tools & Software: MATLAB, EasyEDA, CST Studio Suite, Tinkercad, Arduino IDE, VS Code, Cisco Packet Tracer, Canva, Microsoft Office (Excel, PowerPoint, Word).

Professional Skills: Hardware/software troubleshooting, root-cause analysis, project coordination & scheduling, cross-functional/team communication, documentation & data accuracy.

Languages: Indonesian (Native), English (Intermediate).

WORK EXPERIENCE

Intern, Corporate Secretariat Division

Oct 2025 – Feb 2026

PT Bank Riau Kepri Syariah — Pekanbaru, Riau, Indonesia

- Designed and developed an IoT-based GPS vehicle tracking hardware system with real-time web-based monitoring for advertising partner vehicles (final project), integrating sensors, connectivity modules, and a backend dashboard.
- Built three internal web-based systems supporting corporate operations — ballroom booking, shareholder management, and unsecured installment loan applications — coordinating requirements and delivery timelines with internal teams.
- Managed data entry and document archiving for sponsorship records, ensuring accuracy and traceability.

Information & Communication Technology (ICT) Intern

Feb 2025 – Jun 2025

PT Bumi Siak Pusako — Pekanbaru, Riau, Indonesia

- Designed and built an RFID-based attendance hardware system using an ESP32 microcontroller with RC522 RFID module and Telegram integration, applying embedded systems and hardware-software integration skills.
- Conducted hardware and software testing and troubleshooting on end-user devices, performing root-cause analysis and repairs to prevent recurring field issues.
- Supported monitoring and preventive maintenance of office network infrastructure and hardware.

PORTOFOLIO PROJECTS

- IoT-Based GPS Vehicle Tracking System — End-to-end hardware and web-based monitoring solution for real-time vehicle location tracking, including sensor integration and data acquisition (final project, PT Bank Riau Kepri Syariah).
- RFID & ESP32-Based Attendance System — Hardware-integrated IoT device using an RC522 RFID module with real-time Telegram notifications, including on-device testing and field validation.
- Microstrip Antenna Design at 2.6 GHz — Designed and simulated an RF antenna in CST Studio Suite.
- ESP32-Based Automatic Gate — Automated boom-gate prototype using a servo motor with ultrasonic and PIR sensors for object/vehicle detection.
- Power Supply Circuit — Designed a circuit producing a fixed 5V output and a variable 0–15V output by converting AC input into DC output.
- ADC Circuit with IC ADC0804 — Built an analog-to-digital converter circuit displaying an 8-bit binary value (max 255) across 8 LEDs, demonstrating signal conversion and measurement fundamentals.
- Line Follower Robot — Autonomous robot using an Op-Amp based circuit with LDR sensors for track-following control logic.
- Website-Based Ballroom Booking System (Menara Dang Merdu) — Web application for digitally managing ballroom bookings; users can view schedules and track request status, while admins manage bookings to avoid scheduling conflicts.
- Website-Based Shareholder Management System — Web application for centrally managing and displaying shareholder data, streamlining recording, management, search, and reporting.
- Website-Based Unsecured Installment Loan System — Web application for managing unsecured loan applications and records, streamlining borrower data, loan applications, and installment information.

CERTIFICATIONS

- CCNA (Cisco Certified Network Associate): Introduction to Networks — Cisco Networking Academy ([Verify on Credly](#))

ORGANIZATIONAL & LEADERSHIP EXPERIENCE

Association of Electronics Telecommunication (AET), Politeknik Caltex Riau

Aug 2024 – May 2025

- Coordinator, Spiritual/Religious Affairs Division (2024–2025) — led a weekly divisional program and coordinated cross-group activities.
- Event Coordinator, Student Day 2024; Member, Research and Technology Division (2023–2024); Instructor, Fiber Optic Workshop 2024.

Cisco Netriders Student Activity Unit, Politeknik Caltex Riau

Jan 2024 – Jul 2024

- Head of Events Division, Sumatera Networking Competition VII — led planning and execution of a Sumatera-wide networking competition, including technical implementation and committee coordination.
- Public Relations Division Member, Internal Cisco Networking Competition 2024.